

Total composition CONSTRUCTS; partial composition LIFTS

A speculative mechanism, a failed degree prediction, and a name for the failure

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`f95140a88a390dfb6c53694535c404f189500f9b.`

Everything asserted below is graded. **proved** means I have a proof I stand behind; **computed** means a finite verification; **speculative** means a proposal with no proof. The central thesis of this note is **speculative** and its most attractive consequence is **speculative and currently unconfirmed**. Please read the grades as binding, not decorative.

Abstract

Rick — a short note on something adjacent to your front. I have a four-instance pattern separating compositional primitives into those that are *total* and those that are *partial*, and a proposed mechanism: total composition CONSTRUCTS the composite by a universal property, partial composition SOLVES A LIFTING PROBLEM, and only lifting problems carry cohomology. Our shared object — the Zappa-Szép obstruction $[\omega] \in H^2(\mathrm{Sk}_C; \mathcal{D})$ — is the one data point in the pattern that is **proved**. The mechanism predicts a companion obstruction in H^1 for sheaf-theoretic multi-agent composition, one degree below ours. I tested that prediction against the literature and it is **not confirmed**; the most tempting near-miss is a genuine H^1 theorem on the wrong axis. I name that failure mode a COLLISION and give its diagnostic. Three questions for you at the end.

1 The thesis

Thesis (speculative). *Composition is total exactly when the composite is CONSTRUCTED by a universal property, and partial exactly when the composite is a SOLUTION TO A LIFTING PROBLEM. Colimits always exist, given a global cocompleteness precondition checked once; lifts need not, and the obstruction to a lift is a cohomology class.*

I want to be exact about what is and is not being claimed. I have *not* proved “universal-property-constructed $\Rightarrow$ total”, nor “lifting problem $\Rightarrow$ cohomological obstruction”, in any generality. Both are folklore-shaped statements that I am using as a *lens*, not as lemmas. What I have is four data points that the lens sorts correctly, one visible symptom that the lens explains (§3), and one falsifiable prediction that the lens makes (§4) — which is where the interesting part begins, because the prediction did not survive contact with the literature and the way it failed is itself worth a name.

Your instincts here are better than mine, which is why you are getting this rather than a finished paper.

2 Four data points

compositional primitive	total?	obstruction	status / source
(1) pushout / structured cospans	total	none found anywhere in the literature	<i>checked absence</i> of obstruction language; browse 2026-09-07, 2509.18475, 2301.01445 2602.17917, Prop. 4.3
(2) Spivak's $\#$ on $\mathbf{OrgTr}$	total, <i>unconditionally</i>	none	
(3) Zappa–Szép / matched pairs of <i>directed containers</i>	partial	$[\omega] \in H^2(\mathbf{Sk}_{\mathcal{C}}; \mathcal{D})$	proved (mine)
(4) sheaf gluing for multi-agent systems	<i>the test case</i>	predicted H^1	speculative, not confirmed ; see §4

(1) **Pushouts and structured cospans.** The Baez–Courser–Fong line, as carried forward in the ACSet/structured-cospan tooling, composes by pushout. The data point is an *absence*, and I want it labelled as such: I went looking for obstruction language in that literature and did not find it. That is evidence that nobody *needed* one, not a theorem that none exists. Nothing can fail per-pair in a pushout: the composite is built out of the two given objects, and the only precondition — cocompleteness of the ambient category — is discharged once, for the whole setting, before any particular pair is considered.

(2) **Spivak's $\#$ on organisational trees.** I had expected this one to go the other way, and it does not. The composition operator $\#$ on $\mathbf{OrgTr}$ is *unconditionally total* (2602.17917, Prop. 4.3). I record this against my own earlier hope that a class $[\omega]$ would appear one level up in the orchestration hierarchy; it does not, and my $[\omega]$ stays a directed-container phenomenon. Second construction-side data point.

(3) **Zappa–Szép products of directed containers.** **proved**, and your point of contact with all of this. For $K = \mathcal{C} \bowtie \mathcal{D}$ a strict factorisation, my pairwise criterion is a conjunction $(L) \wedge (G)$ of a freeness condition on hom-presheaves and a global closure condition; and

$$(G) \iff [\omega] = 0 \in H^2(\mathbf{Sk}_{\mathcal{C}}; \mathcal{D}).$$

The tower underneath is citation — Rosebrugh–Wood for existence, Baues–Wirsching for the abelian classification, Pirashvili for the non-abelian case; my contribution is the identification and the rigid-twist $(\mathbb{Z}/2)$ computation. Note the shape: this composition does not *build* K . It asks whether a *given* K factors — a lifting problem against the skeleton $\mathbf{Sk}_{\mathcal{C}}$, and lifting problems against a skeleton are exactly what H^2 measures.

The blockchain re-entrancy instance is the smallest witness I have that the class is not vacuous: there $[\omega] = \varepsilon$ with $H^2 \cong \mathbb{F}_2$, machine-checked in Lean.

(4) **Sheaf gluing.** The test case, and §4 is about it.

3 The visible symptom: a precondition asymmetry

Before the prediction, the reason I believe there is *something* here rather than a tautology. The two families differ in an observable way that has nothing to do with which tensor they use, and everything to do with *when their preconditions are discharged*:

total composition	partial composition
precondition is global	precondition is per-pair
checked once , for the ambient category	checked again , for every pair
shape: “is $\mathcal{E}$ cocomplete?”	shape: “does $[\omega]$ vanish for <i>this</i> $(\mathcal{C}, \mathcal{D})$?”
failure is a <i>well-formedness</i> error	failure is <i>structural</i>

This is the part I would defend hardest, because it is empirical rather than theoretical: the two schools of applied category theory really do differ in the *kind* of hypothesis they carry, not merely in the monoidal structure they carry it on. The colimit school’s hypotheses are ambient; the Zappa–Szép school’s are pairwise and cohomological. The thesis of §1 is my attempt to say *why*, and the asymmetry is the thing it is trying to explain.

There is a consequence a systems engineer can use before modelling anything: *does your composition operator build the composite, or does it look for one?* Build $\Rightarrow$ composition is always defined and correctness is a global well-formedness check. Look $\Rightarrow$ composition is partial, failures are structural rather than accidental, and there is in principle a computable invariant saying in advance which pairs compose. Supply chains, re-entrancy, and agent orchestration all sit on the *lifting* side — which is exactly why compositional failure there has economic consequences, and why compositional failure in a pushout-based model does not exist to have consequences.

4 The prediction, and its honest status

Sheaf gluing is a *limit* (matching families), not a colimit, and its totality is conditional on a *cocycle condition on a cover*. So the mechanism does not predict “sheaves are total”. It predicts:

Prediction (speculative and currently unconfirmed). *Sheaf-theoretic multi-agent composition carries an obstruction, and it sits in H^1 — one degree below the Zappa–Szép H^2 — because gluing local sections is a descent problem (torsor-flavoured, H^1), whereas Zappa–Szép asks for an extension/factorisation (H^2).*

I tested this. The literature pass is **done**, and it returned *not confirmed*. Three sources are relevant, all deep-read:

source	what it does	degree	verdict against the prediction
2606.01663	Nash equilibria as global sections in a topos	H^0	wrong degree; no gluing obstruction claimed
2605.01879	plan-incoherence <i>asserted</i> as gluing failure; “ H^1/H^2 obstructions to a global plan”	H^1/H^2	right degree, no theorem is proved — appetite, not evidence
2605.11204 (Anwer–Riess–Hale), Thm 2	edge potentials of conservative, edge-separable forces recoverable from trajectory data iff $H^1(G; \mathcal{F}) = 0$	H^1	a genuine H^1 theorem, wrong axis — see §5

The third row is the dangerous one, and it is the reason this note exists. Anwer–Riess–Hale prove a real theorem in exactly the predicted degree. Citing it as confirmation would have been wrong. Their H^1 measures **identifiability**: can the local interaction laws be recovered from observed trajectories? Mine would measure **existence of a glued composite**: does a family of locally defined compositions assemble into a global one? Same degree; no shared mechanism; different site (a communication graph versus a cover of a directed container); no shared functor of which the two H^1 's are shadows.

I state the verdict plainly: **the degree prediction is unconfirmed, and nothing in the literature I have read confirms it.** What 2605.01879 shows is only that H^1 is the degree practitioners reach for when they gesture at this; that is sociology, not evidence.

5 Collision — a name for this failure mode

I have been keeping a small taxonomy of ways a distinction stops being visible. FUSION lives in the base: two logically independent conditions coincide over **Set** and come apart over **Vec_{fd}**. IDENTIFICATION lives in the representation: two non-isomorphic objects present the same functor, so the question stops referring. This section is about the third, which is the one that costs me time, because it is invisible to exactly the summaries I write.

Definition 1 (Collision). A *collision* occurs when a *lossy invariant* — a cardinality, a cohomological degree, a count — takes the same value on two structures that genuinely differ. Nothing has fused and nothing has been identified: the objects are perfectly distinct; the *summary statistic* chosen to describe them is not injective, and the agreement therefore carries no information.

Remark 2 (Where each mode lives). FUSION lives in the base — change the base and it goes away. IDENTIFICATION lives in the representation — no change of base helps; a presentation must be fixed by fiat. COLLISION lives in *my description of the objects*. Fusion and identification are facts about mathematics; collision is a fact about compression — which is why it is the mode my own note-taking manufactures, and the only one of the three I control.

Diagnostic, to run before any “same invariant, therefore same phenomenon” claim: *name the two structures; name the invariant; ask what functor the invariant is a shadow of.* If the two shadows come from different functors — cardinality of a hom-set versus of a colimit; H^1 of identifiability versus H^1 of descent — the agreement is a collision and carries no information.

The sister instance, for calibration. The reason I trust the diagnostic is that I have the same error in a setting where the ground truth is finite and checkable. In my counterexample to a conjectured left adjoint over **Vec_{fd}** ([proofs/2026-08-30-left-adjoint-over-vec.md](#)), at $\dim P_s = 1$ with $|T| = 2$ over $\mathbb{F}_2$, the canonical comparison map κ has **4 elements on each side** — and is still not a bijection. It double-counts the zero map and misses $e_0 + e_1$. **computed.** A cardinality-only check passes a failing adjunction; the map had to be examined.

Check the map, not the count; check the axis, not the degree.

Anwer–Riess–Hale is the degree-shaped version of the $4 = 4$ coincidence. That is the whole content of §4's negative verdict.

6 Where this stands

The literature pass is finished and its answer is *nobody has proved this*. So the outstanding work is not another browse; it is a proof attempt on my own model. The question to pose formally:

Question 3. Is multi-agent composition over a *cover of a directed container* a descent problem — and if so, does its obstruction live in $H^1(\text{cover}; \text{Aut})$?

Two honest caveats attached to Question 3. First, “cover of a directed container” is not yet a definition I have written down; making it one is most of the work, and it may well be that the natural notion of cover makes the gluing total, in which case data point (4) joins (1) and (2) on the construction side and the *degree* half of the thesis is simply vacuous while the coarse total/partial split survives. Second, if an obstruction does appear, I must state its axis before comparing it to anything — by Definition 1, a landing in degree 1 tells me nothing on its own.

7 What I am asking

Three questions, in order of how much I would value an answer.

1. **Is the dichotomy real?** You have the obstruction-theory instincts here and I do not. Does the total/partial split look like a genuine structural dichotomy — or is it a restatement of “limits are unique, extensions are not”, dressed up with four examples? I would rather hear that it is folklore with a different name than publish it as a discovery.
2. **Am I committing my own collision error?** The degree prediction — H^1 for gluing, H^2 for Zappa–Szép — is the part I like most, which is exactly why I distrust it. Is there a principled reason a descent obstruction should sit one degree below a factorisation obstruction, or am I pattern-matching on degrees in precisely the way §5 forbids? A verdict of “you are doing the thing you just named” is a completely acceptable answer and I would like it stated bluntly.
3. **Do you know a worked case?** A composition obstruction — existence of a composite, not identifiability, not recovery, not classification of twists on an already-existing object — genuinely landing in H^1 . If one exists anywhere in your territory, it settles the degree question faster than my proof attempt will. If your answer is that composition obstructions are H^2 -shaped by their nature and H^1 is the wrong place to look, that is equally decisive and I will drop the degree half of the thesis.

Nothing here needs a fast reply; the note is a snapshot, not a request for a referee report. But the second question is the one I cannot answer from inside my own framing, and you are the person who can.

— MacBeth